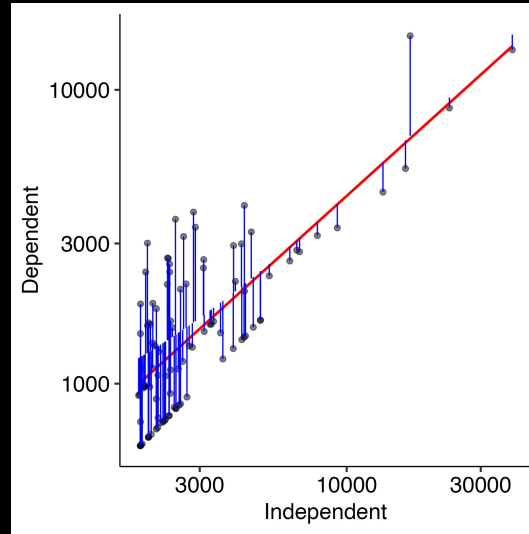


Linear Regression - II



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Introduction to Public Health Informatics

DH 302

Lecture 11 || Wednesday, 12th February 2025

Are the following models linear?

$$y = ax + b$$

$$y = ax^2 + b$$

$$y = ab + x$$

$$y = e^x \beta + c$$

$$y = e^\beta x + c$$

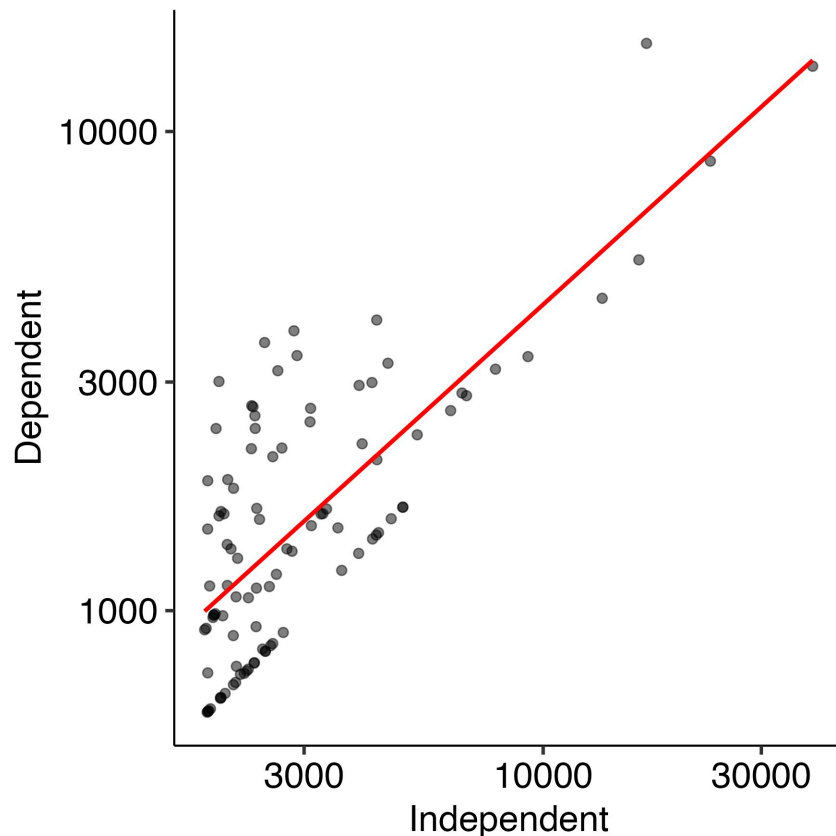
The “best fit” line

Given observations $\{\mathbf{x}_i, y_i\}_{i=1}^n$ in ordinary least squares (OLS) regression, we estimate the parameters β . Two possible approaches:

- Loss function based optimization $\rightarrow$ minimize the “prediction” error
- Distributional way: Assume the data is coming from a distribution and model it explicitly

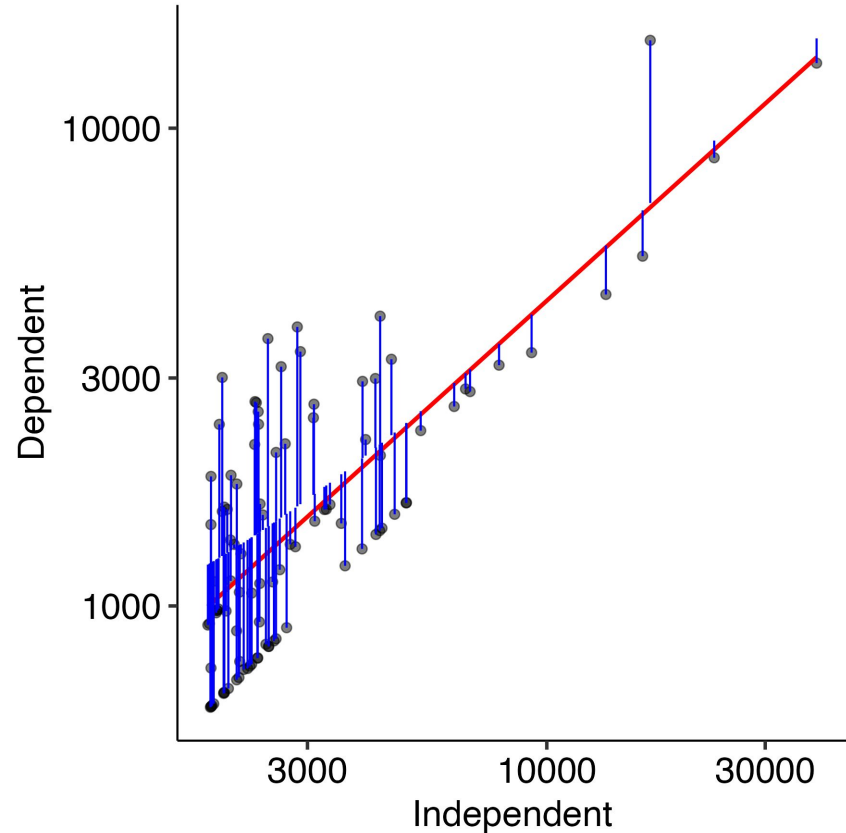
Linear regression: The Goal

- “Regression analysis” = Problem of estimating relationship between a **dependent variable** and one or more **independent variables**
- “Linear regression” = Simplest example of regression analysis where the relationship between the dependent and the independent variable is assumed to be linear in the form of parameters



Linear regression: How is the line fit?

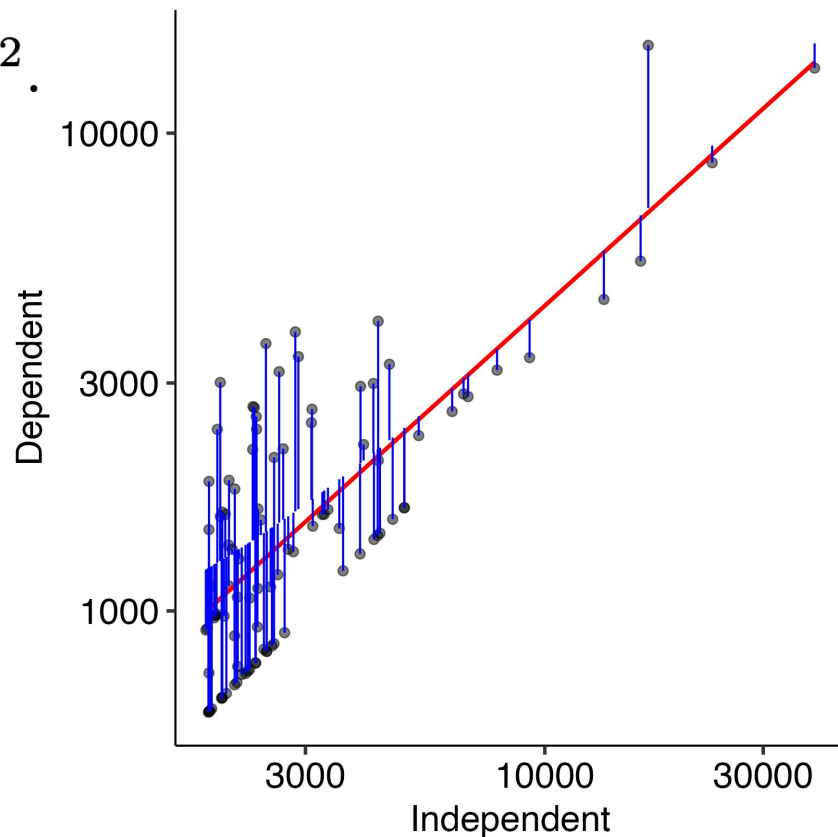
- **Goal:** Fit a line of the form $y = mx + b$.
 - M = Slope of the line
 - b = Intercept
- **Approach:** We find a line such that the sum of square distances (vertical) between the actual points and the line is minimized. Why **square**?
- Squared distances are “easier” to optimize



Fitting the line: Method 1 (multivariable calculus)

Minimize $f(m, b) = \sum_{i=1}^n (y_i - (mx_i + b))^2.$

- Take partial derivatives w.r.t m and b and set to zero
- Having a square here is helpful as the resulting derivative is easy (say compared to if it was an absolute value)



Fitting the line: Method 2 (linear algebra)

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{p,n} \end{bmatrix}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \quad \hat{\mathbf{y}} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{pmatrix} = \mathbf{X}\boldsymbol{\beta}$$

Minimize

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

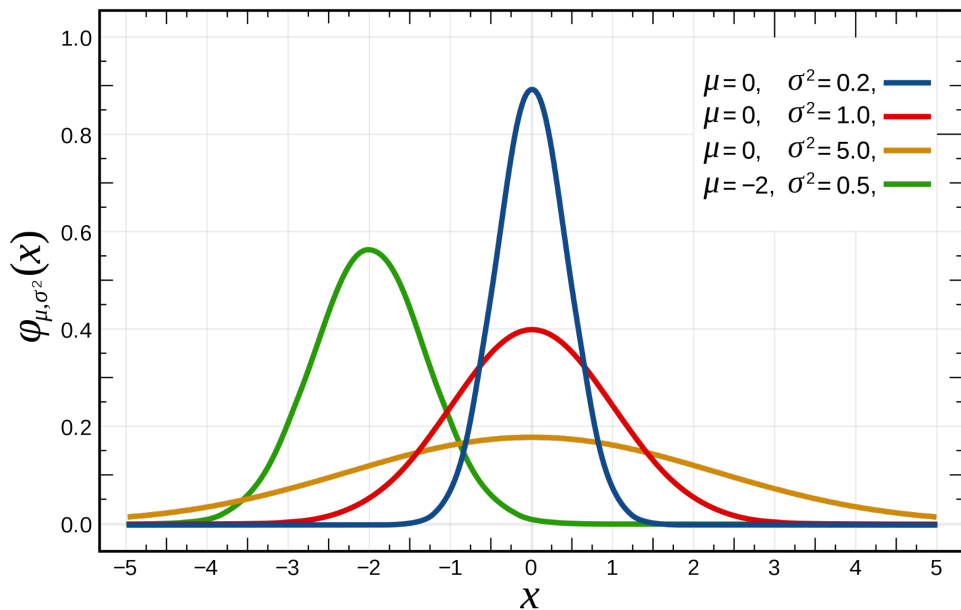
- $\mathbf{Y} = (y_1, \dots, y_n)^T$, is a point in an n -dimensional vector space
- Least squares optimization problem = Finding the nearest point on a subspace spanned by a column matrix $\mathbf{X}$ defined from the dependent variables.
- Invoking orthogonality, the solution emerges as $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$.

Fitting the line: Method 2 (Probabilistic)

- Normal (gaussian) distribution:

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}.$$

- Two parameters:
 - μ = mean
 - σ^2 = variance
- “Standard normal” has $\mu = 0, \sigma^2 = 1^2$.



Fitting the line: Method 2 (Probabilistic)

Independent variables (x) are measured without error, but there is error in the measurement of dependent variables (y)

$$y_i = mx_i + b + \epsilon_i$$

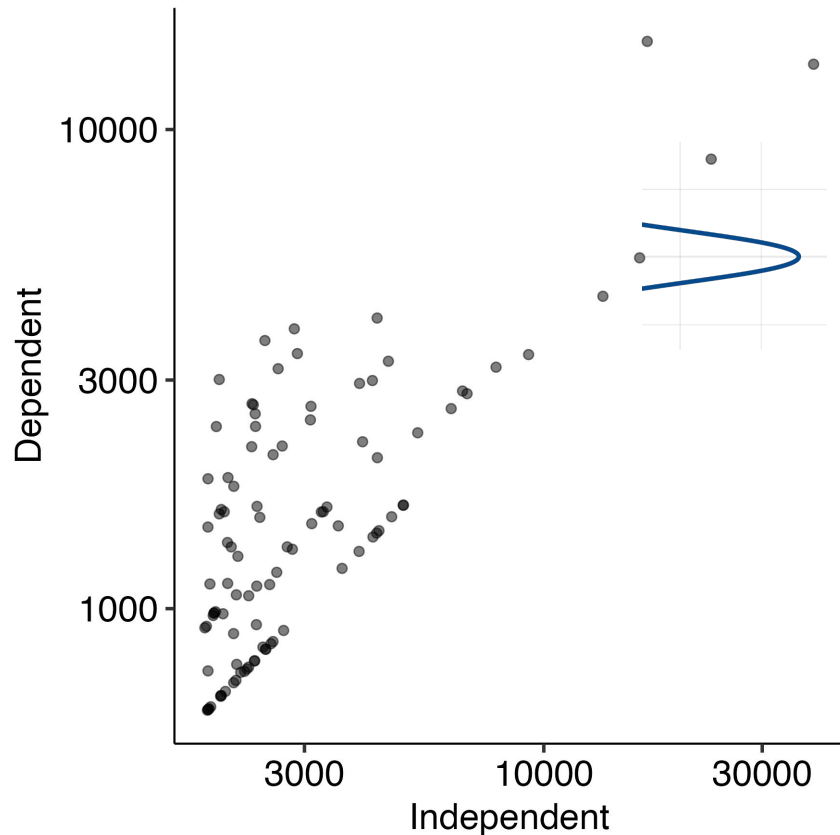
$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- Probability of observing y_i is then given by:

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}.$$

- The likelihood of observing data $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ is given by:

$$\mathcal{L}(m, b) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}.$$



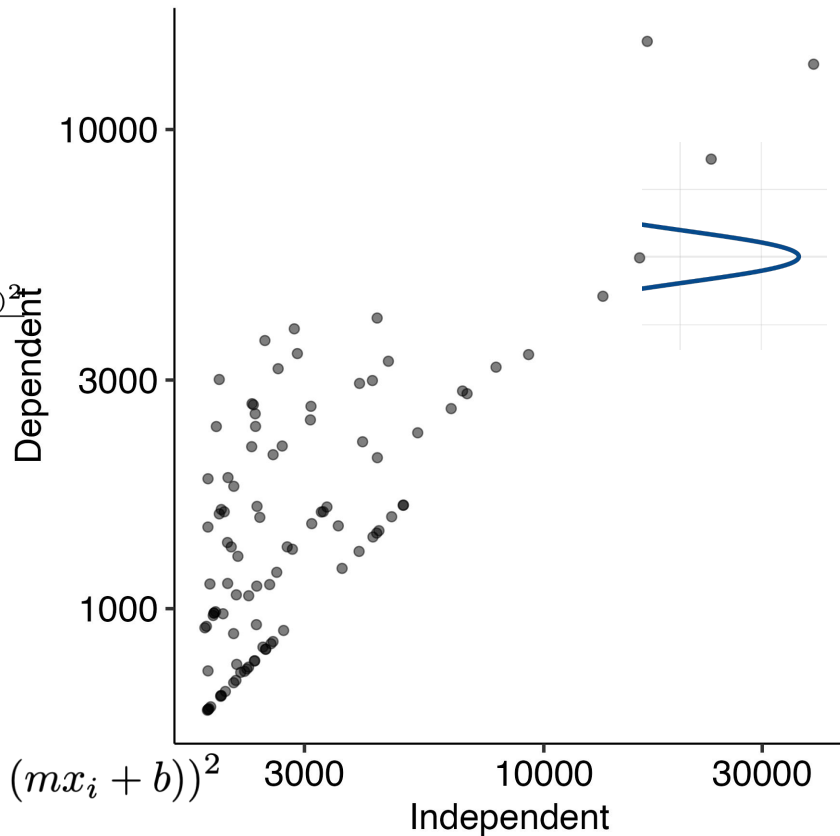
Fitting the line: Method 2 (Probabilistic)

- To find the “best fit” line, we would want to maximize the likelihood of observing the given data given the parameters

$$\operatorname{argmax}_{m,b} \mathcal{L}(m,b) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}$$

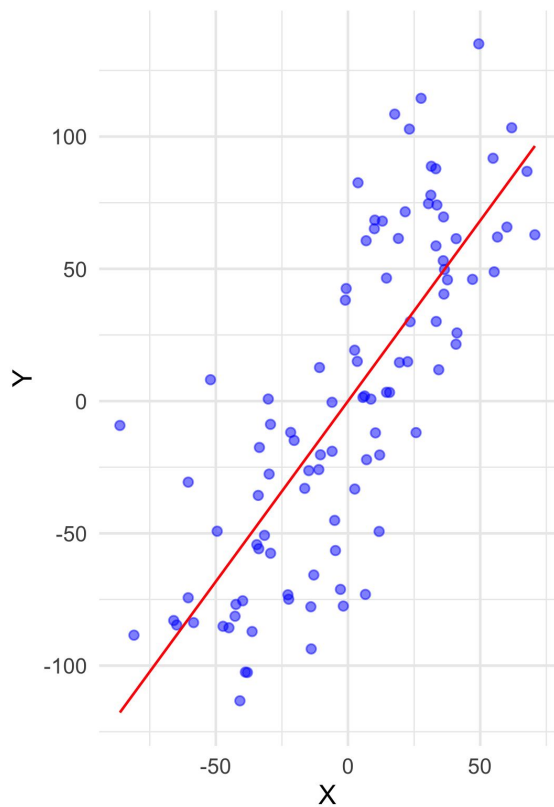
- Take the logarithm which is a monotonically increasing function and find the parameters m , b that maximize the logarithm of the likelihood function

$$\begin{aligned} \operatorname{argmax}_{m,b} \log(\mathcal{L}(m,b)) &= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (mx_i + b))^2 \\ &= \operatorname{argmin}_{m,b} \sum_{i=1}^n (y_i - (mx_i + b))^2. \end{aligned}$$

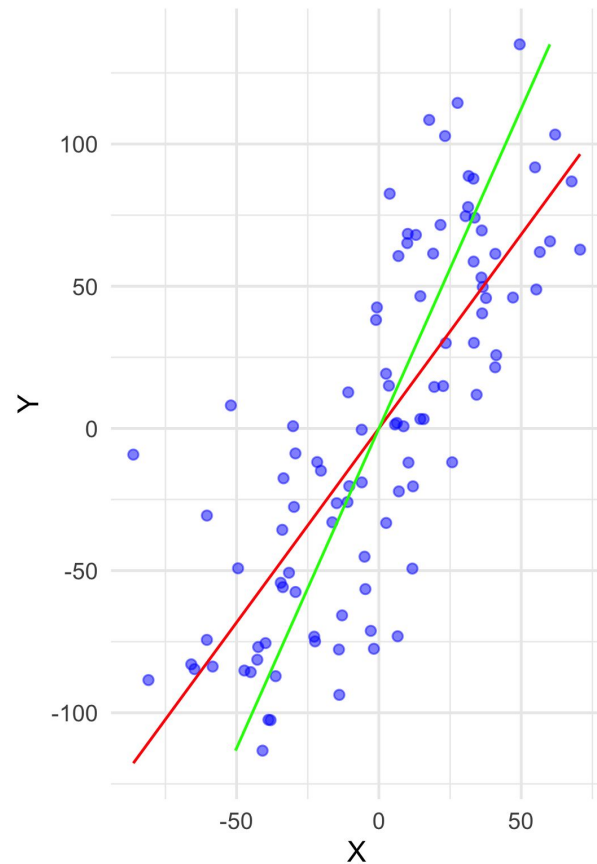
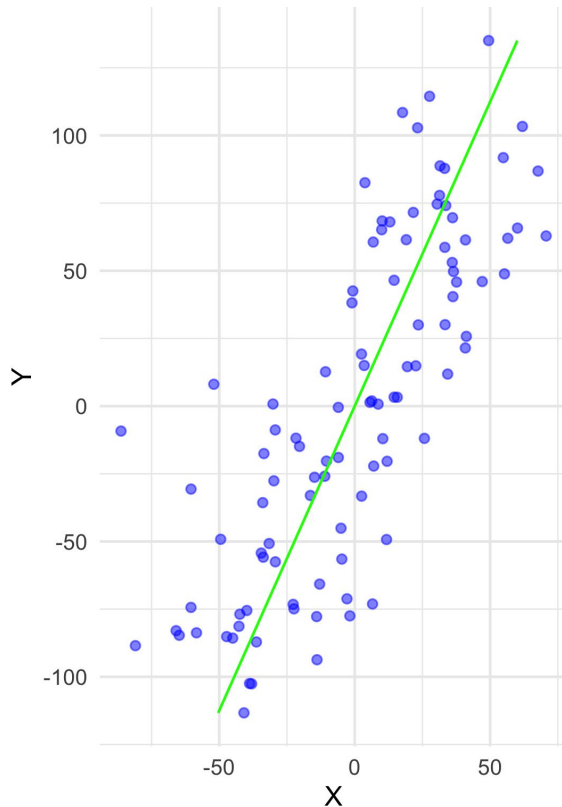


Dependent and independent variables should not be interchanged

`lm (y~x)`

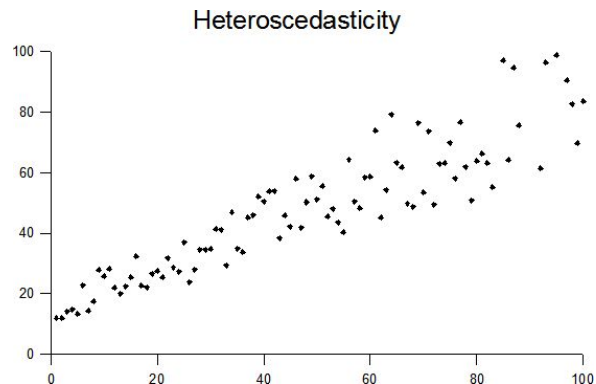
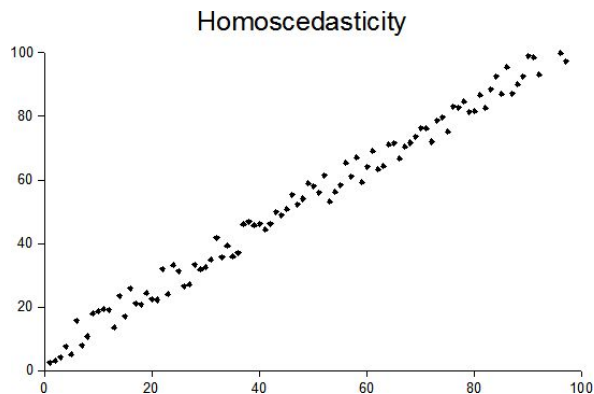


`lm (x~y)`



Homoskedasticity vs Heteroscedasticity

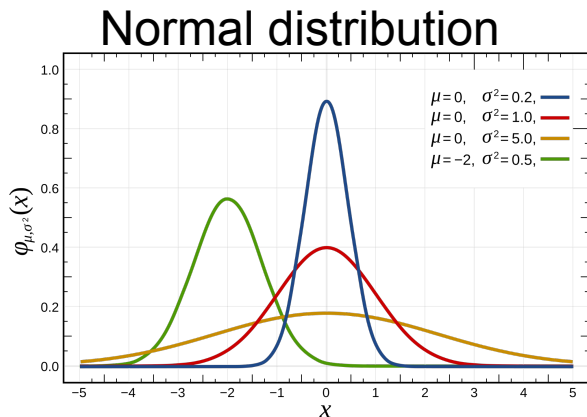
An inherent assumption in fitting a linear model is that of **homoscedasticity** → variance is constant across observations and is not a function of the observed quantity



What is so special about squares?

$$f(m, b) = \sum_{i=1}^n (y_i - (mx_i + b))^2$$

Least squared error fit can be interpreted as maximum likelihood for a linear model with gaussian noise in observations

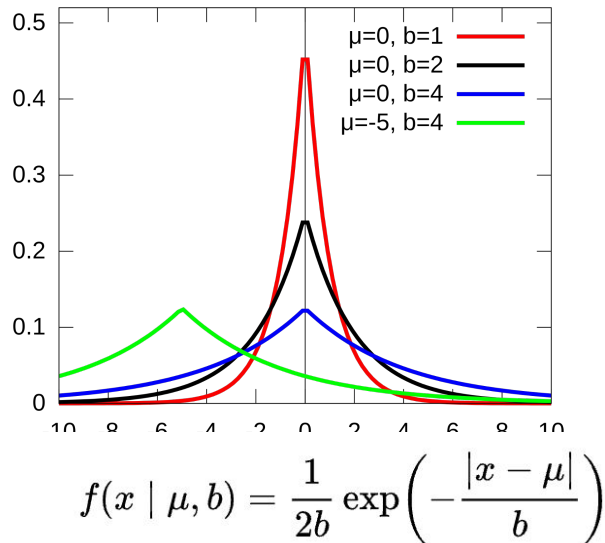


$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f(m, b) = \sum_{i=1}^n |(y_i - (mx_i + b))|$$

Absolute fit can be interpreted as maximum likelihood for a linear model with laplacian noise in observations

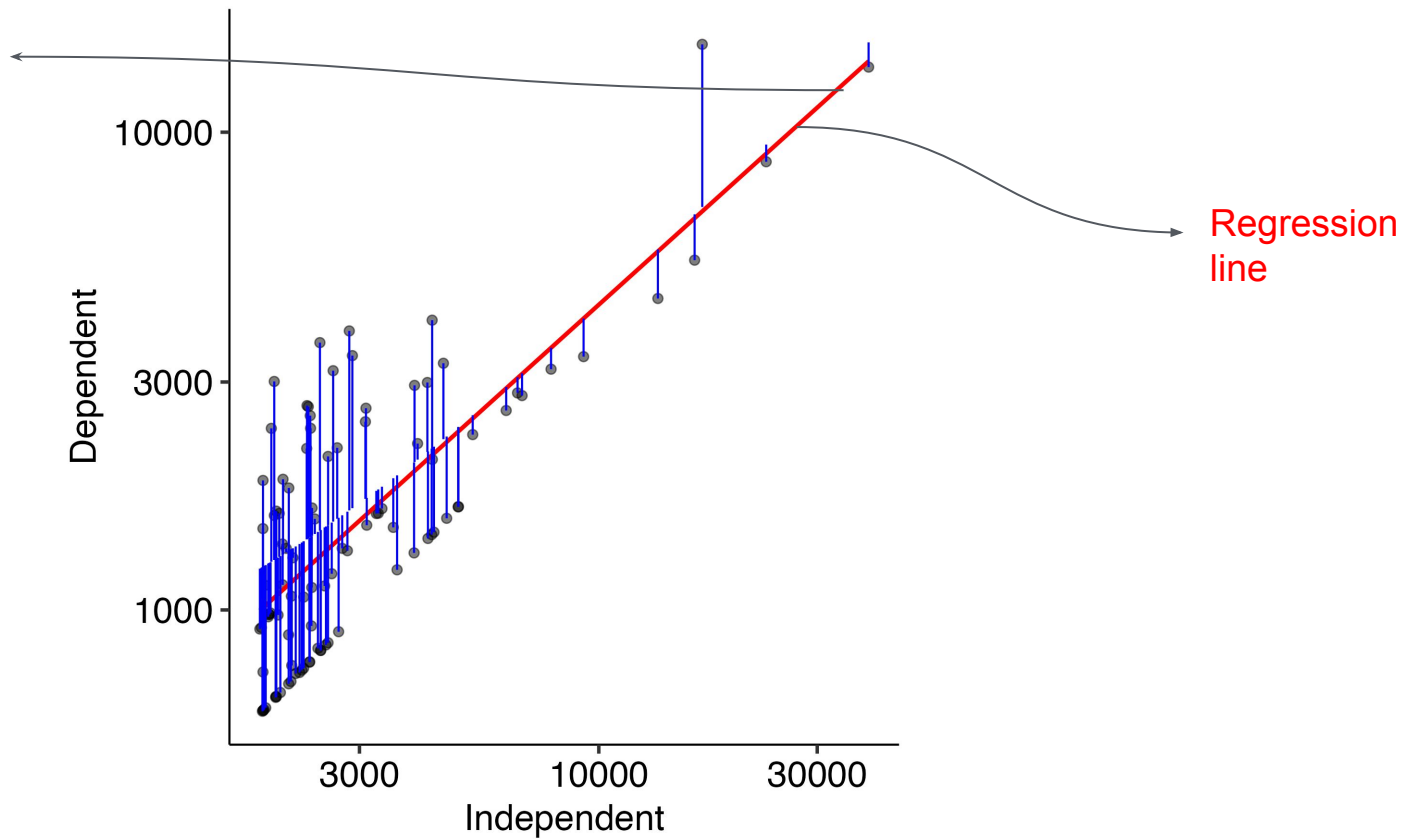
Laplace distribution



$$f(x | \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

The residuals

Residuals



The residuals

Observation pairs

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Estimate

$$\hat{\beta}_0 \quad \hat{\beta}_1$$

$$y_i \approx \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Prediction

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Residual

$$e_i = y_i - \hat{y}_i$$

Minimize residual
sum of squares

$$\text{RSS} = e_1^2 + e_2^2 + \dots + e_n^2,$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

Coefficient
estimates

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i \text{ and } \bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$$

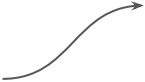
Estimating the accuracy of the coefficient estimates

$$Y = \beta_0 + \beta_1 X + \epsilon. \rightarrow \text{Defines population regression line}$$

β_0 Intercept = Expected value of Y when X is 0

β_1 Slope = Average increase in Y for a unit increase in X

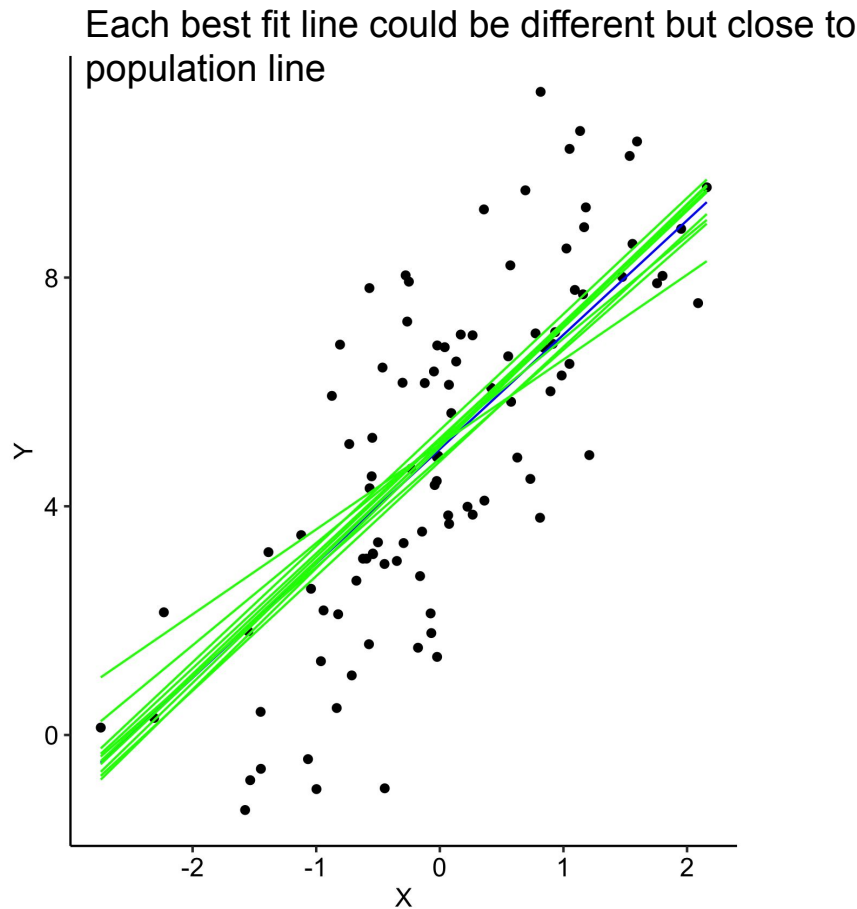
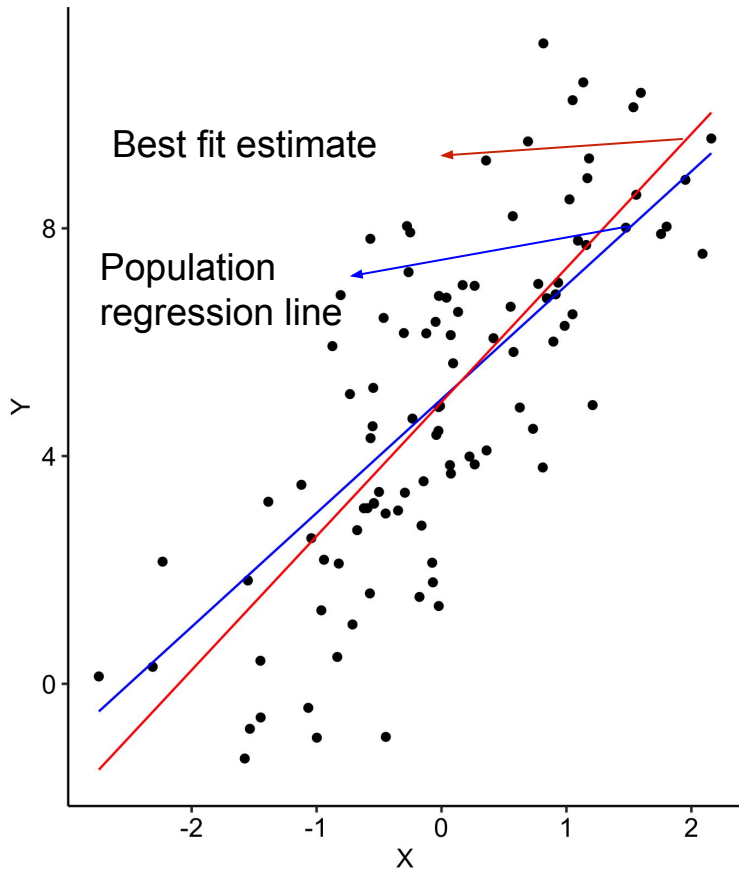
ϵ A catch-all term $\rightarrow$ true relationship between X and Y is not linear or there are other covariates but this is supposed to cat

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$


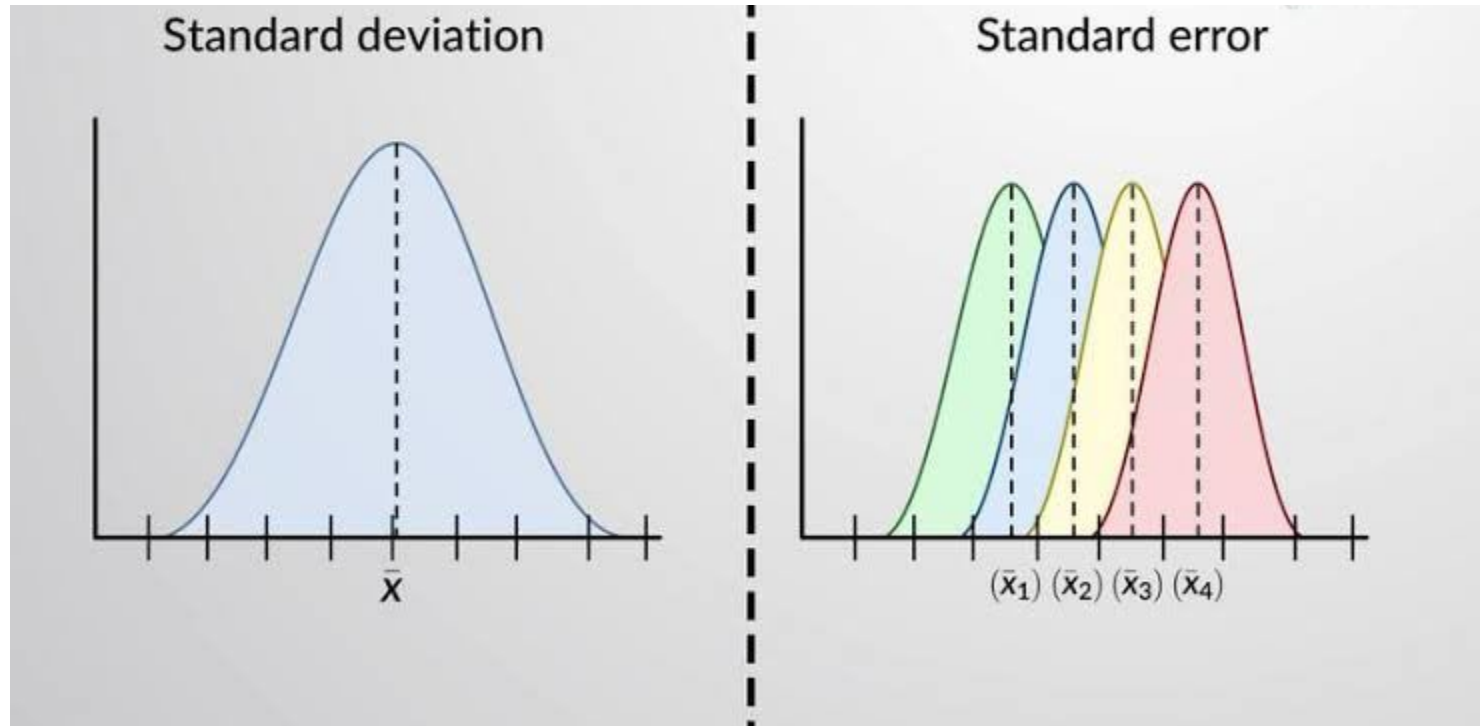
Estimates of the population parameters

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

The residuals



Review: Standard error of the mean



How close are the estimates β_0 and β_1 to the true population values

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

$$\sigma^2 = \text{Var}(\epsilon)$$

$$\text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

σ^2 is unknown so estimated from the data in the form of residual standard error

$$\text{RSE} = \sqrt{\frac{1}{n-2} \text{RSS}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Confidence intervals

- Standard errors can be used to estimate confidence intervals
- A 95 % confidence interval is defined as a range of values such that with 95 % probability, the range will contain the true unknown value of the parameter
- In other words, if you were to repeat your sampling experiment 100 times (obtaining (x,y)) and reestimate the parameters of linear model, ~95 times the estimates would lie in your estimated confidence interval
- 95% CI for β_1 $\left[\hat{\beta}_1 - 2 \cdot \text{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \text{SE}(\hat{\beta}_1) \right]$
- 95% CI for β_0 $\hat{\beta}_0 \pm 2 \cdot \text{SE}(\hat{\beta}_0).$

What is the linear model testing?

H_0 : There is no relationship between X and Y $H_0 : \beta_1 = 0$

H_a : There is some relationship between X and Y $H_a : \beta_1 \neq 0$,

If $\beta_1 = 0$ the model reduces to $Y = \beta_0 + \epsilon$, $\rightarrow$ X is not associated with Y

- To test if β_1 is different from 0, we need to test how different is it from 0
- How different β_1 from 0 depends on the value of β_1 but also how confident are we that β_1 takes that value
- Confidence is quantified using the standard error $SE(\hat{\beta}_1)$ of β_1

Test statistic for β_1 : $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$ t follows a t-distribution with n-2 degrees of freedom

Assessing the fit using R^2

- RSE provides an absolute measure of lack of fit of the model to the data
- RSE is measured in the units of Y, it is not always clear what constitutes a good RSE
- R^2 provides a quantitative fit of the model = what proportion of variance is explained by the linear model

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{TSS} = \sum (y_i - \bar{y})^2$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

r^2 and R^2

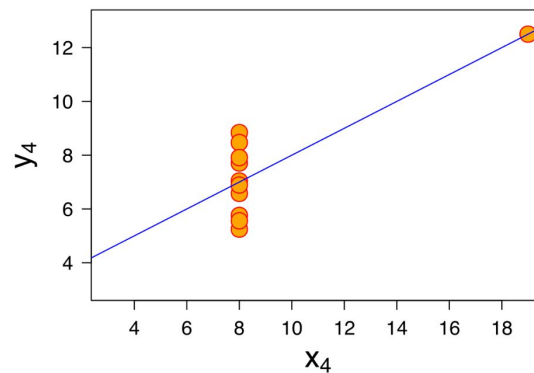
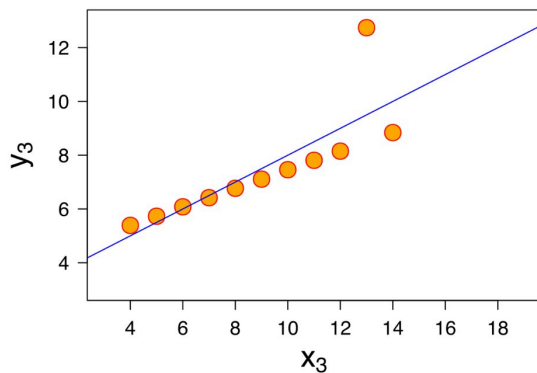
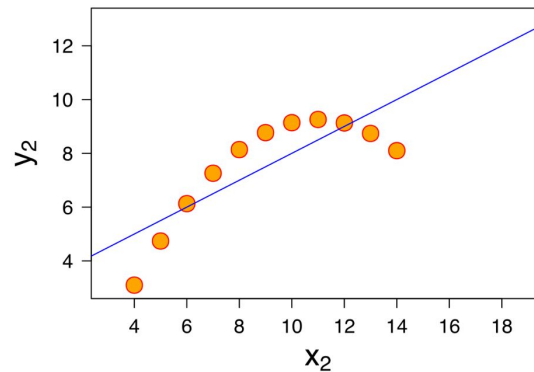
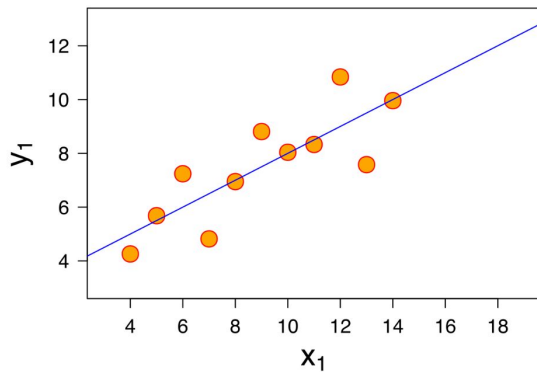
For “simple” linear regression (with one independent variable), the proportion of variance explained by the linear model is equal to the square of correlation coefficient

$$r = \text{Cor}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

$$R^2 = r^2$$

Metrics aren't everything, why visualization is important

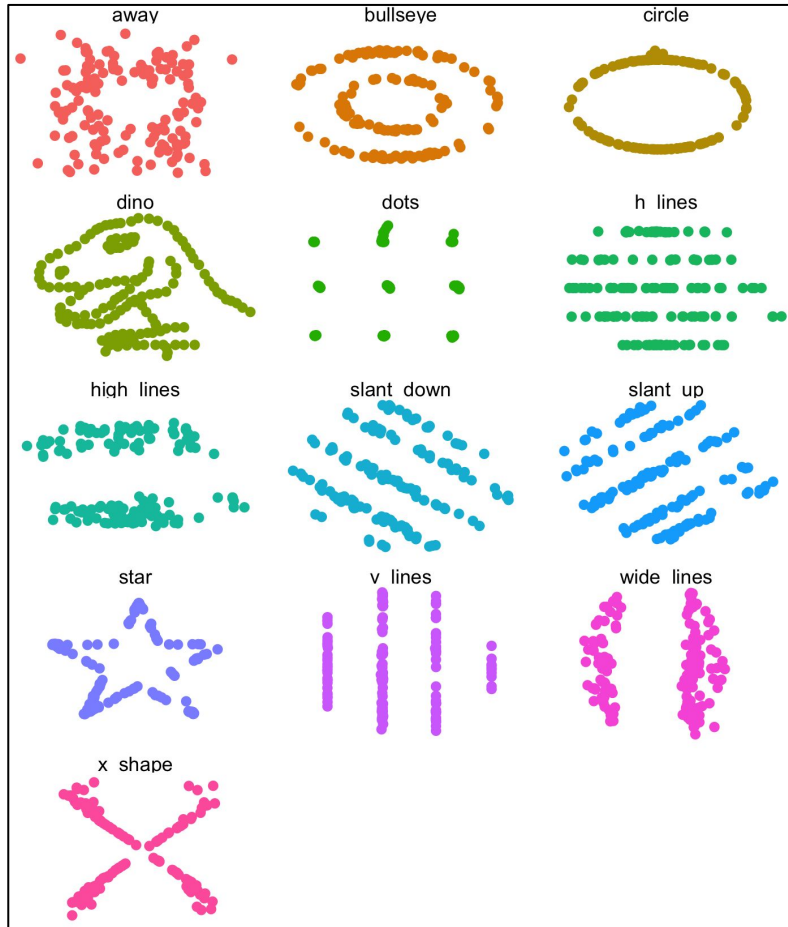
Anscombe's quartet: Same summary statistics, different relationship



[Anscombe's quartet](#)

[Source](#)

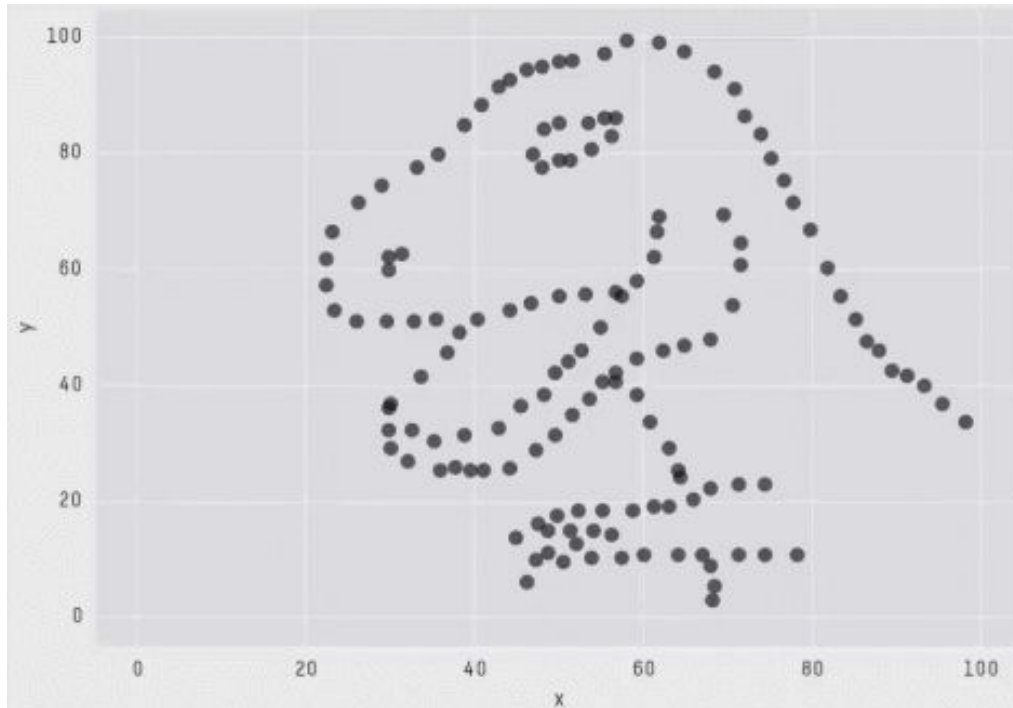
Datasaurus dozen



13 datasets with same mean and std.
deviation upto 2 decimal places

**Always visualize your data before starting
to pump out numbers/metrics**

Datasaurus dozen

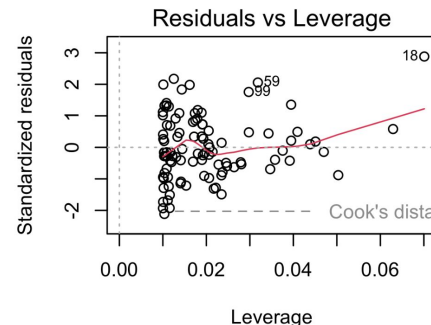
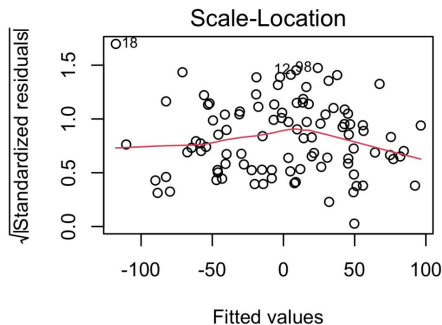
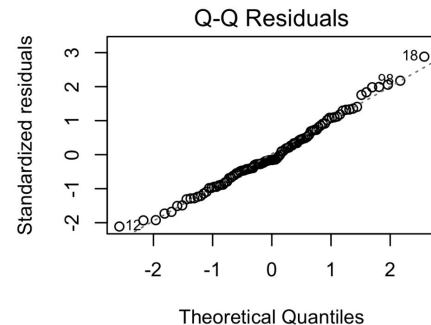
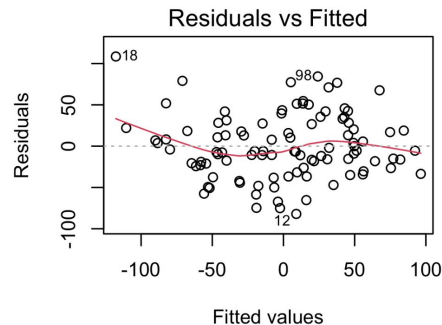


X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526

Diagnosing the fit

- *Residuals vs Fitted* - Do residuals have non-linear patterns?
 - Good: Equal spread across fitted values
 - Bad: Some distinctive pattern (increasing/decreasing/U-shaped)
- *Q-Q* - Are the residuals normally distributed?
 - Good: Straight line (normality assumption is satisfied)
 - Bad: Strong deviations from normality
- *Scale Location* - are residuals spread evenly along ranges of predictors (check for homoscedasticity)
 - Good: Equal spread across fitted values
 - Bad: Some distinctive pattern (increasing/decreasing/U-shaped)
- *Residual vs Leverage* - Identify “influential” cases:
 - Good: Most points in $[-2,2]$ of y axis (standardized residuals)

```
linear_fit <- lm(y~x, data = as.data.frame(X))  
par(mfrow = c(2, 2))  
plot(linear_fit)
```



Running linear regression in R and diagnostics

Cook's Distance: Used to measure influence of a data point. How different would predictions be if the i-th point were to be deleted?

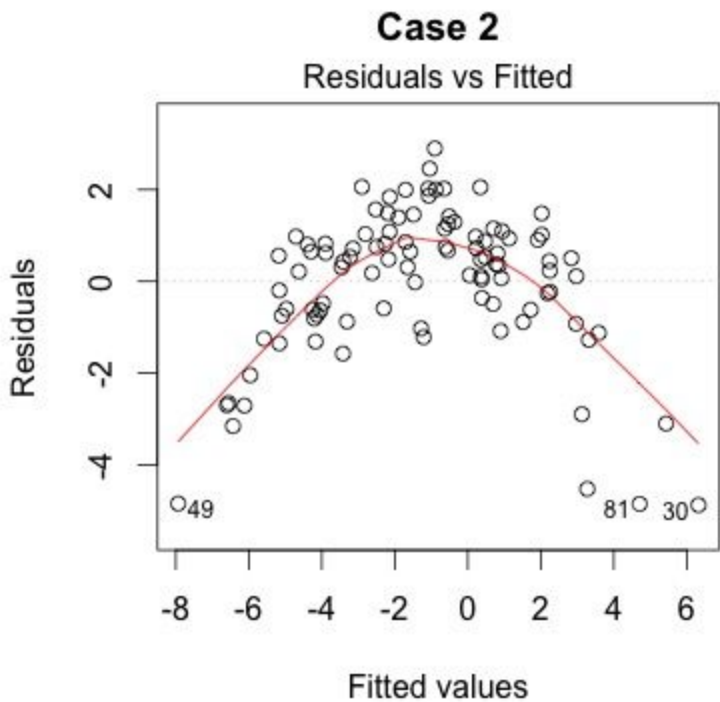
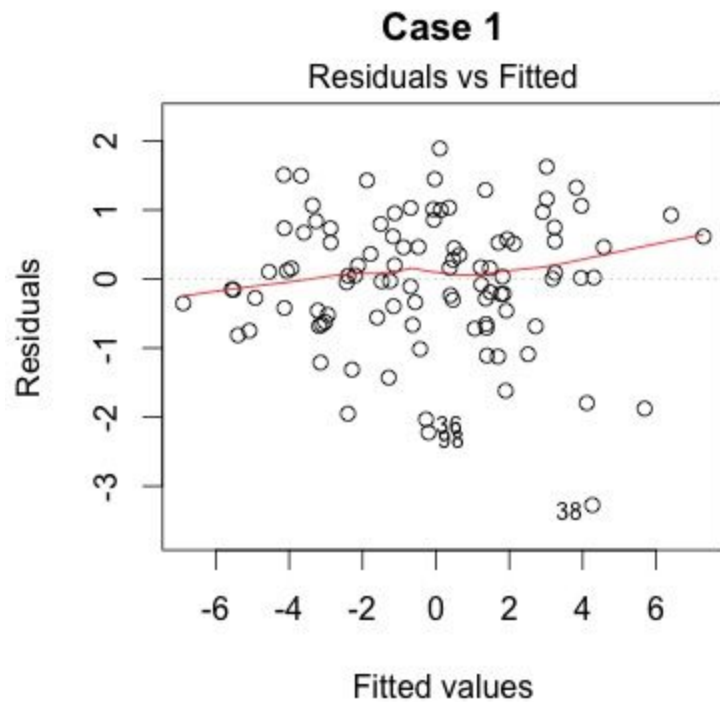
$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{ps^2}$$

P = rank of model

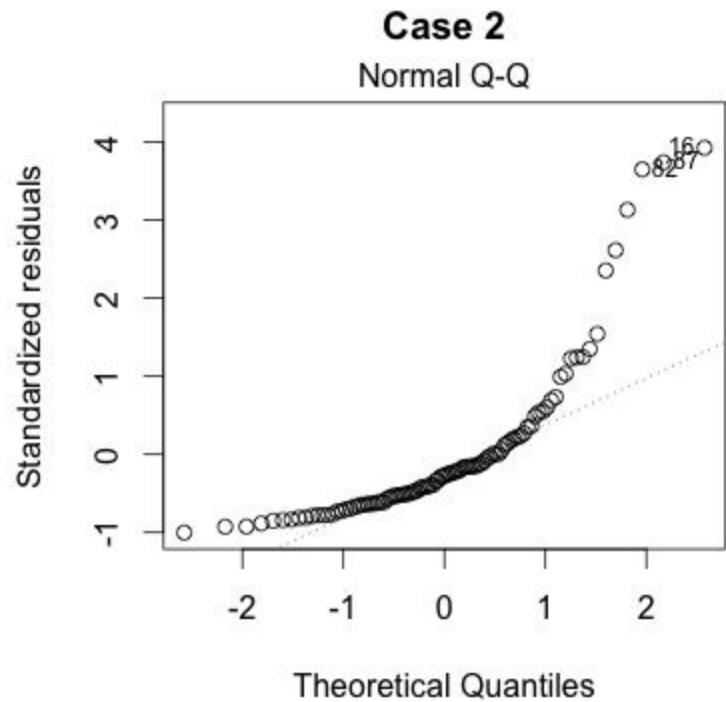
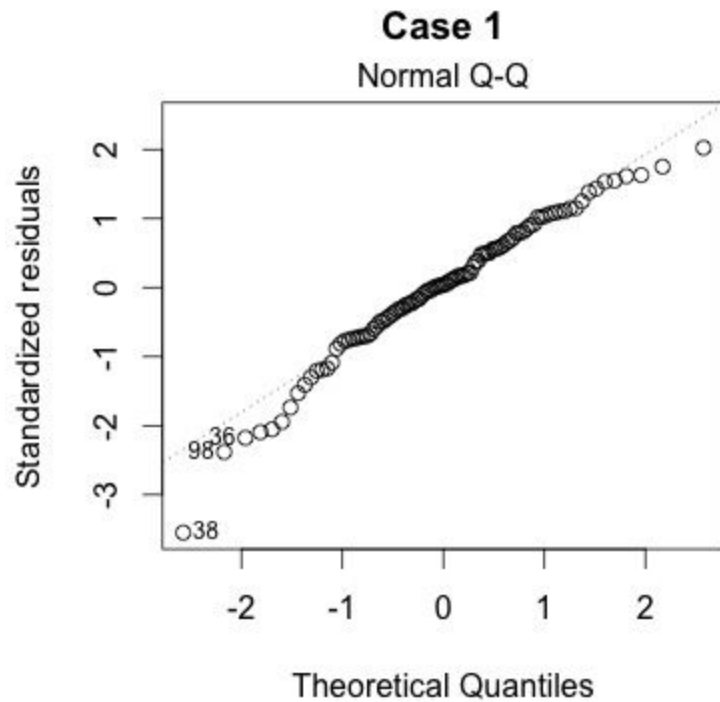
s² = Squared error of the model

$\hat{y}_{j(i)}$ Fitted y value when i-th observation is deleted

The good vs the bad



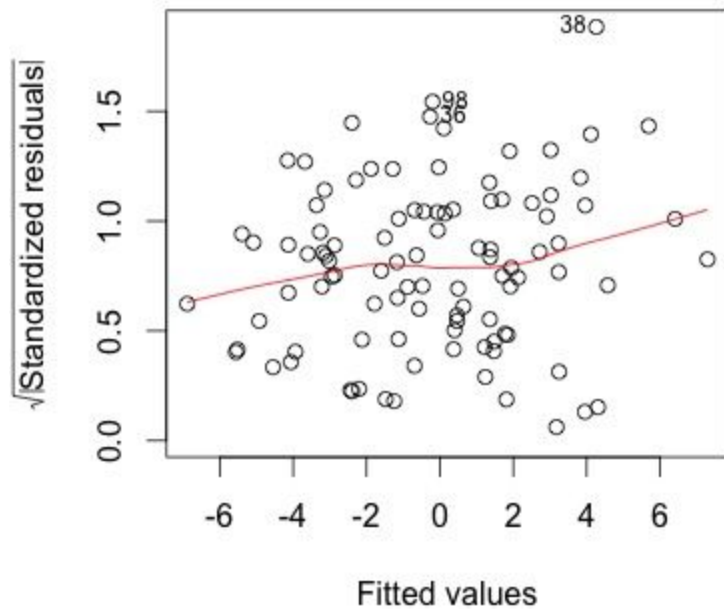
The good vs the bad



The good vs the bad

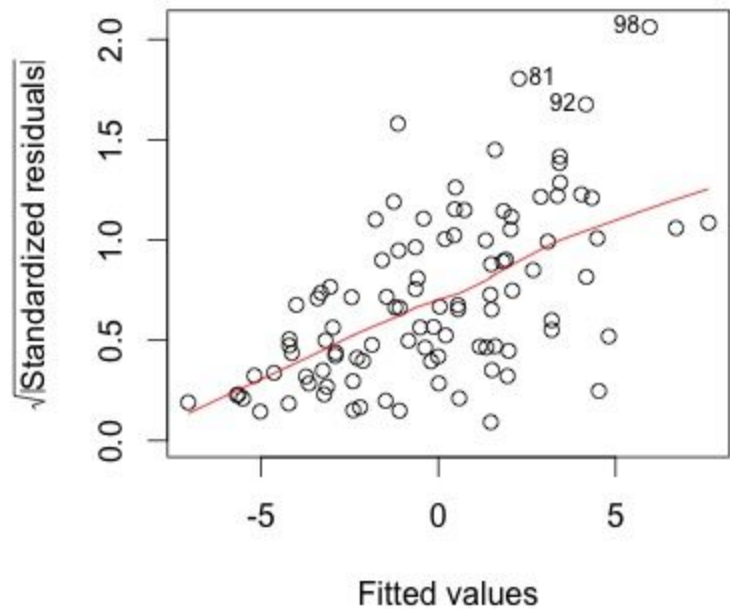
Case 1

Scale-Location

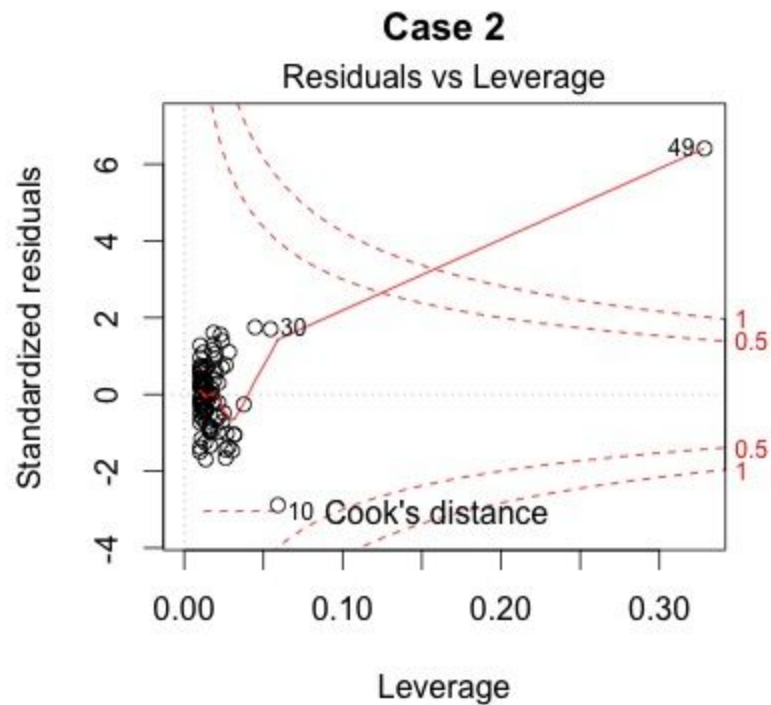
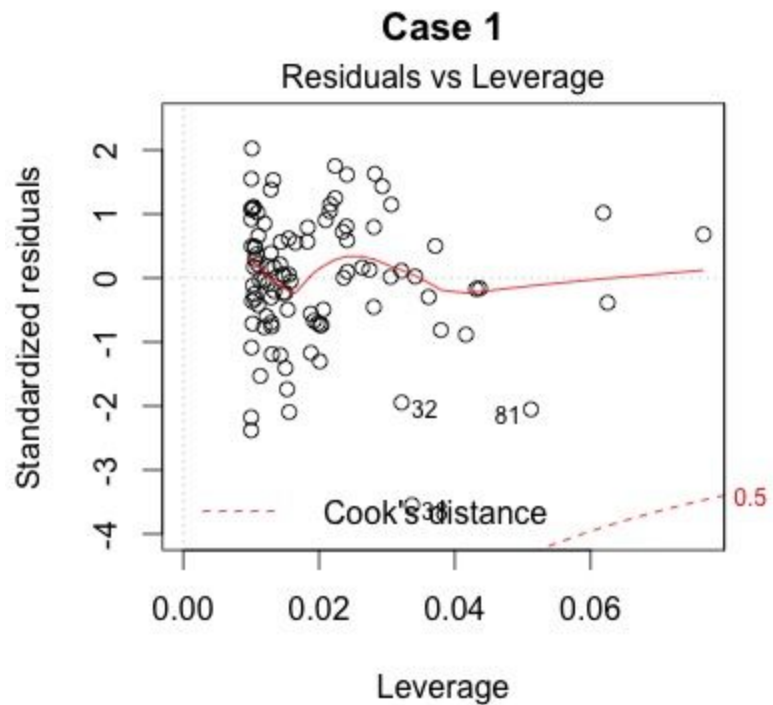


Case 2

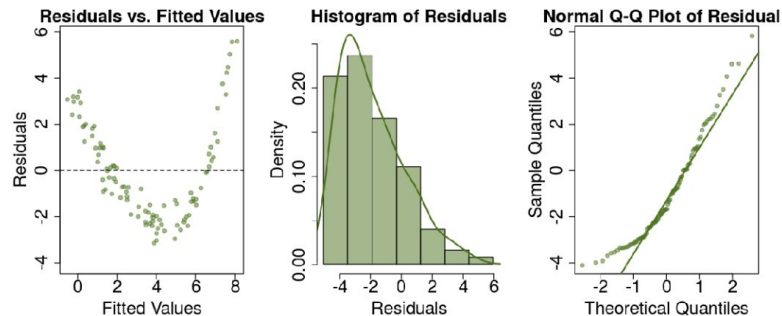
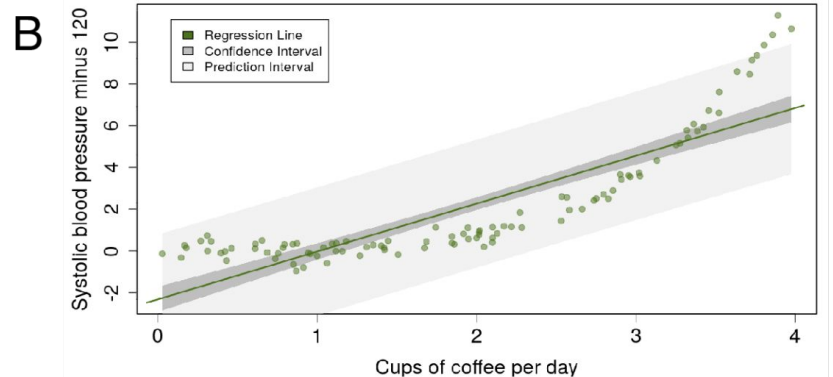
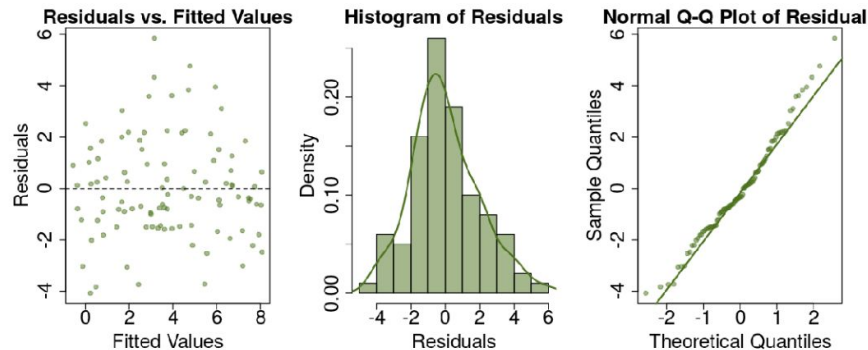
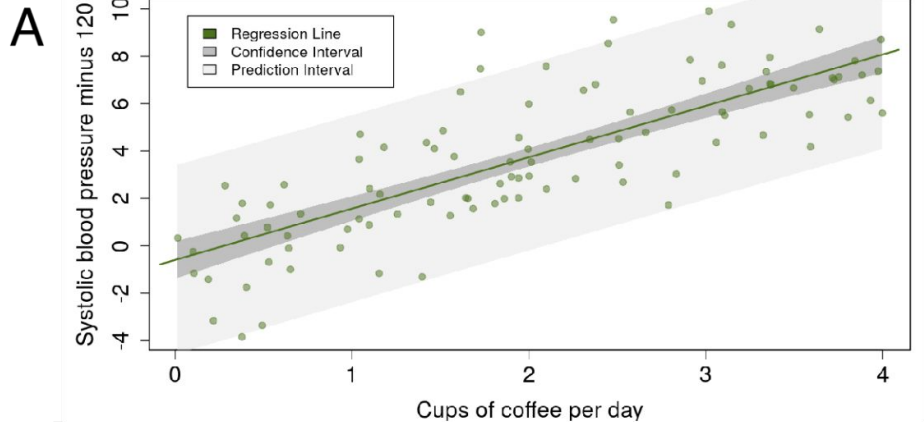
Scale-Location



The good vs the bad



The good vs the bad



[Source](#)

